

Supplementary Information

Place-field repetition is what a predictive map must do when self-motion cannot resolve it: a cognitive map represents space only up to the symmetries self-motion cannot break

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This document collects supporting figures for the main text. All panels are generated directly from the result files by `project5_symmetry/analysis/make_paper_figures.py` and `rate_maps.py`; no values are entered by hand.

1 Place-field rate maps, all extracted units

The main-text place-cell figure shows a small number of example units per condition. To show that this selection is representative rather than chosen, the following figures give the full set of extracted units for each condition, ranked by Skaggs spatial information and left otherwise unselected. Each tile is one unit's mean firing-rate map over the arena, smoothed for display and normalised to its own peak.

C_1 , full HD: 40 most spatially informative units

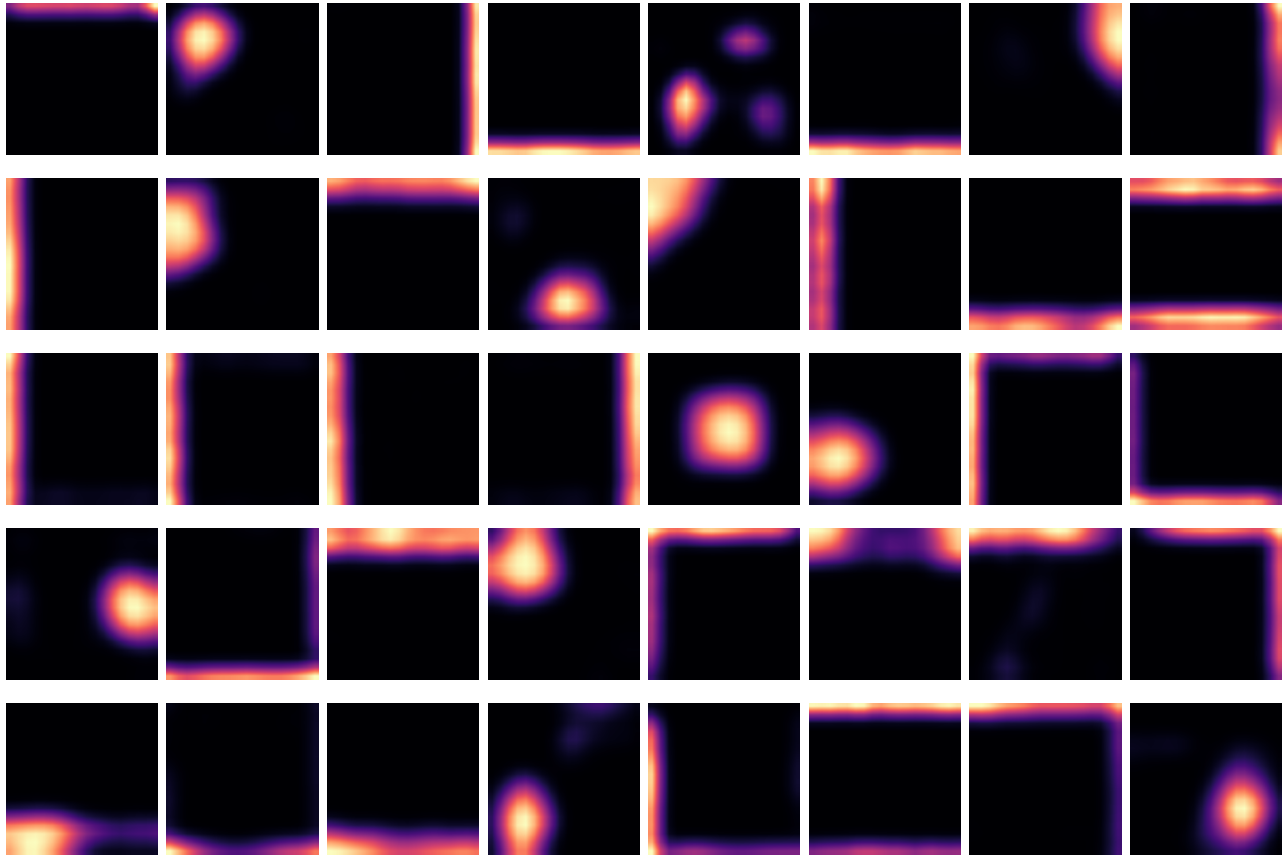


Figure S1. Asymmetric arena, full head-direction encoding. Units carry single, sharp place fields.

C_2 , full HD: 40 most spatially informative units

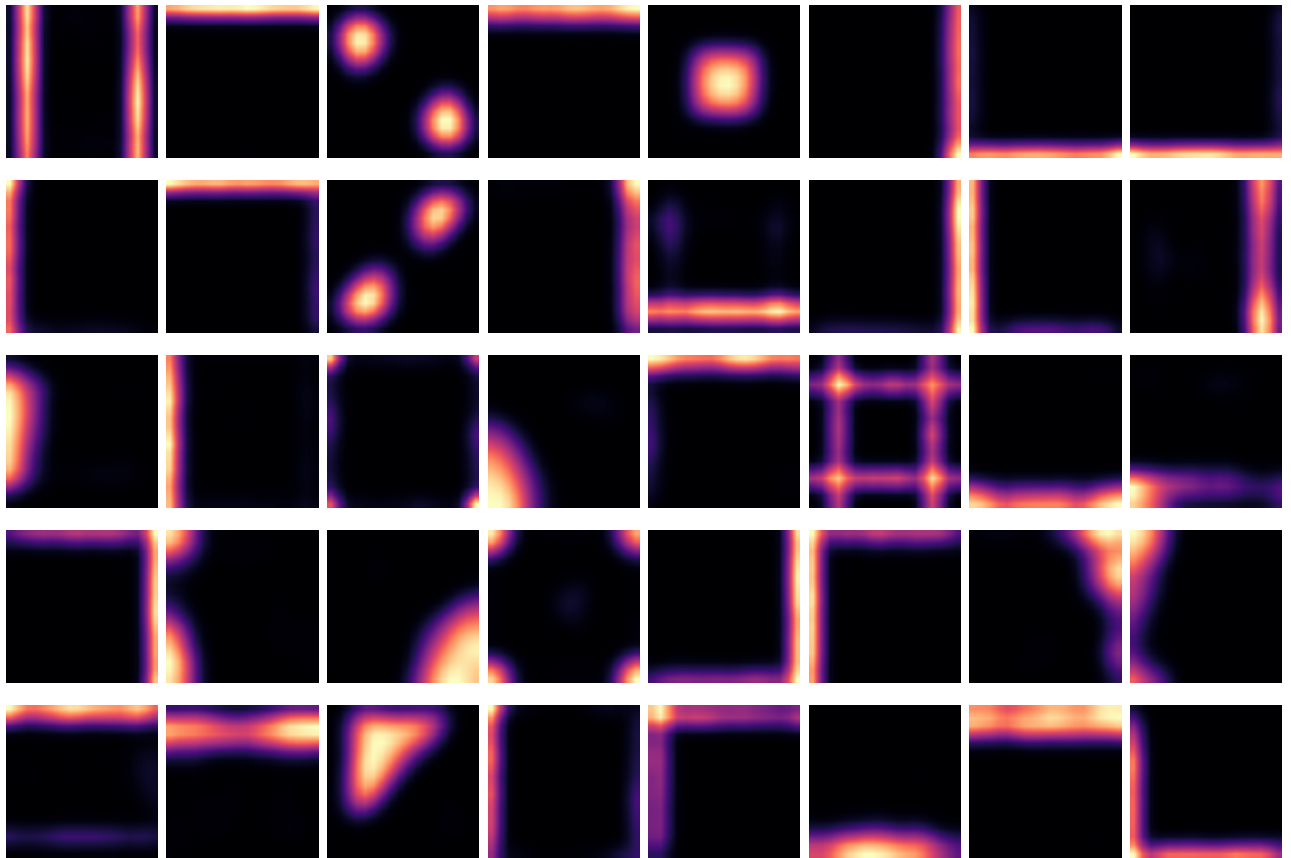


Figure S2. C_2 arena, full head-direction encoding. With the full compass available the fields do not systematically repeat.

C_2 , axis HD: 40 most spatially informative units

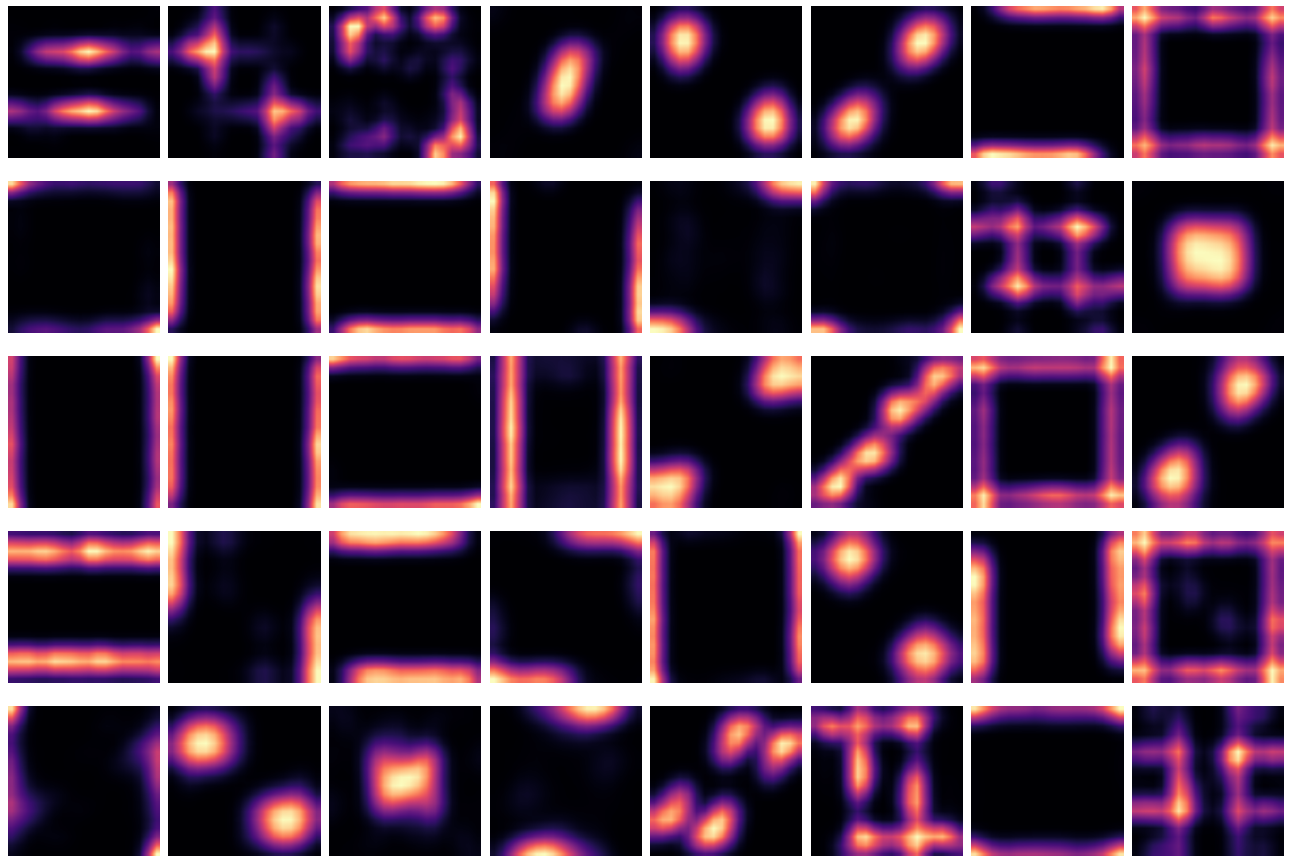


Figure S3. C_2 arena, *axis* encoding. The encoding is invariant under the 180° rotation, and fields repeat at 180° -related locations across the population.

C_4 , const HD: 40 most spatially informative units

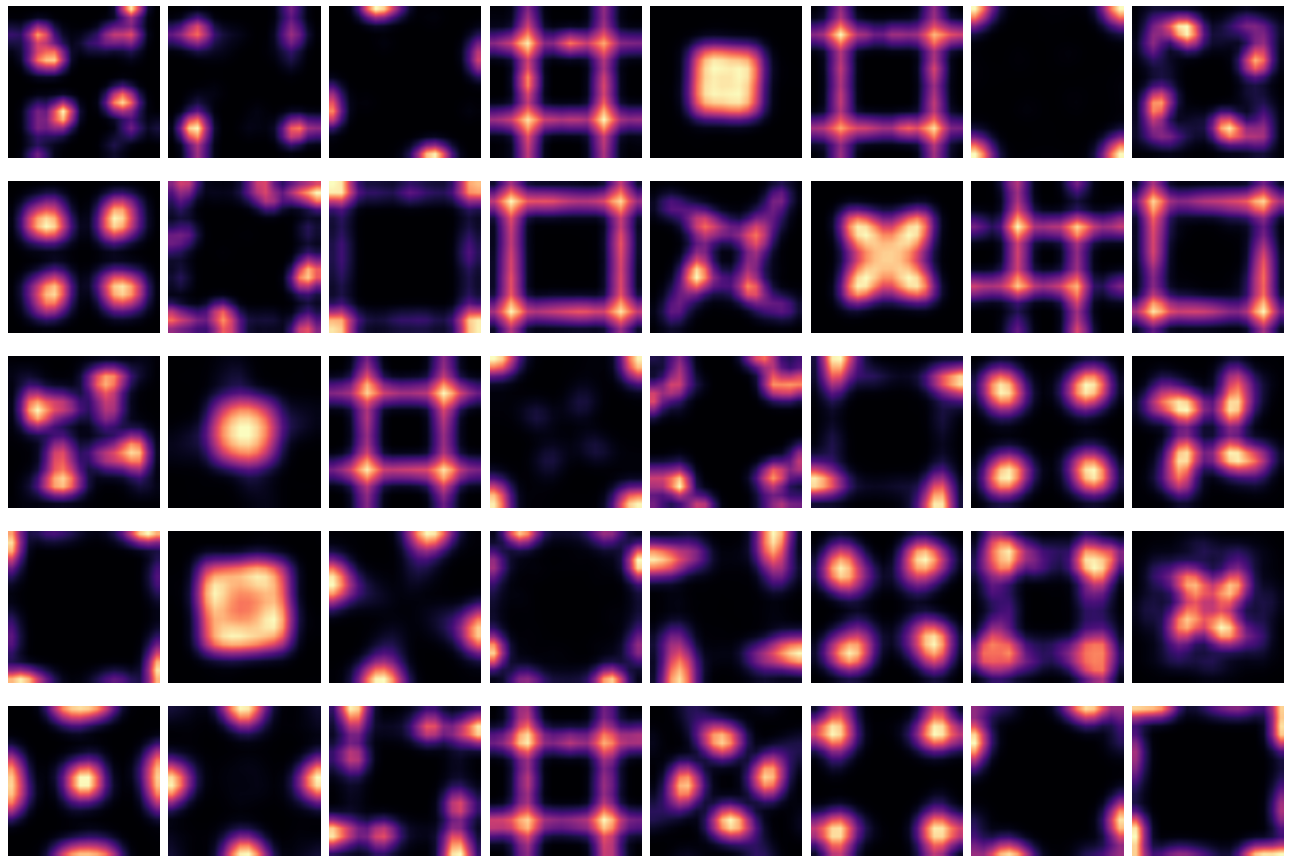


Figure S4. C_4 arena, *const* encoding. The encoding is invariant under the full C_4 group, and fields repeat at all four 90° -related locations.

2 Cell-type composition

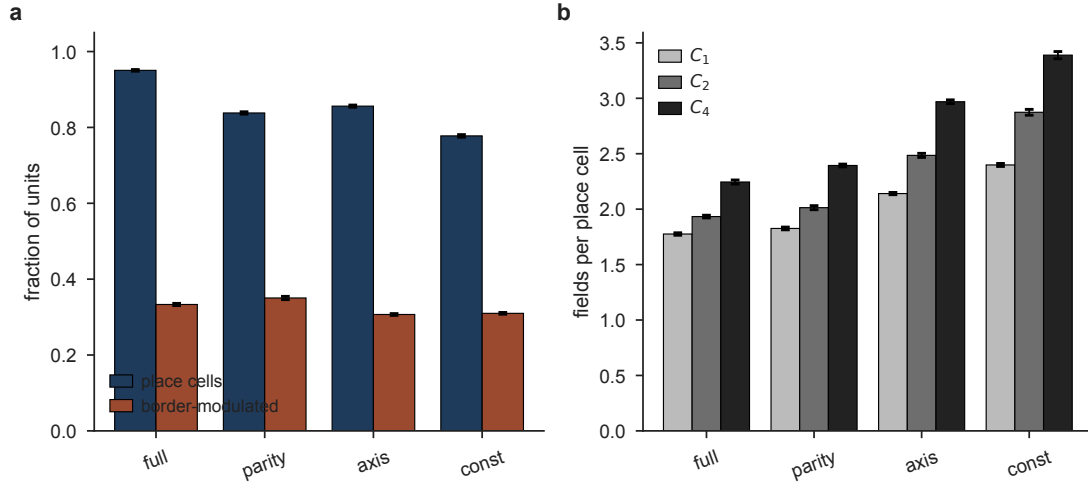


Figure S5. The trained population is a realistic cell-type mixture, scored by the criteria applied to real recordings. **(a)** Fraction of units classified as place cells (Skaggs spatial information above threshold) and as border-modulated (a third or more of the rate-map mass in the outer ring), by head-direction encoding, pooled across arenas. The composition is stable across encodings. **(b)** Fields per place cell by encoding and arena ($C_1/C_2/C_4$): it rises with both encoding invariance and arena symmetry, from 1.78 to 3.39, the multi-field signature of the fold. Folding rearranges where fields sit and multiplies them without changing the cell-type makeup. Bars are mean $\pm$ s.e.m. over networks.

3 Population manifolds of the topology arenas

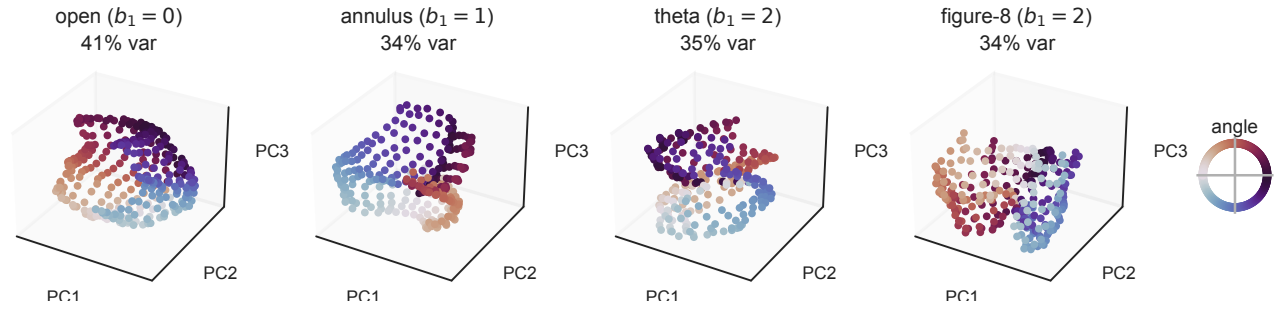


Figure S6. Position-conditioned population manifolds for the four topology arenas, PCA to three dimensions (one point per passable cell, coloured by angle around the arena centre). Position is mapped smoothly in every arena, but the manifolds are high-dimensional: three principal components capture only 34–41% of the variance, and no low-dimensional loop is visible even for the annulus ($b_1 = 1$) or the theta and figure-8 arenas ($b_1 = 2$). This is consistent with the persistent-homology analysis, which does not recover the arenas' loop topology (main text, Limitations): the code is metrically organised well before, and without ever cleanly acquiring, the arena's topology.

4 Diffusion-map geometry of the manifolds

Diffusion maps embed a point cloud through the eigenfunctions of a Markov random walk built on it, so the embedding respects diffusion distance — a density-robust, geometry-aware metric — and the eigenvalue spectrum reads out intrinsic dimension. Both structures survive this nonlinear readout.

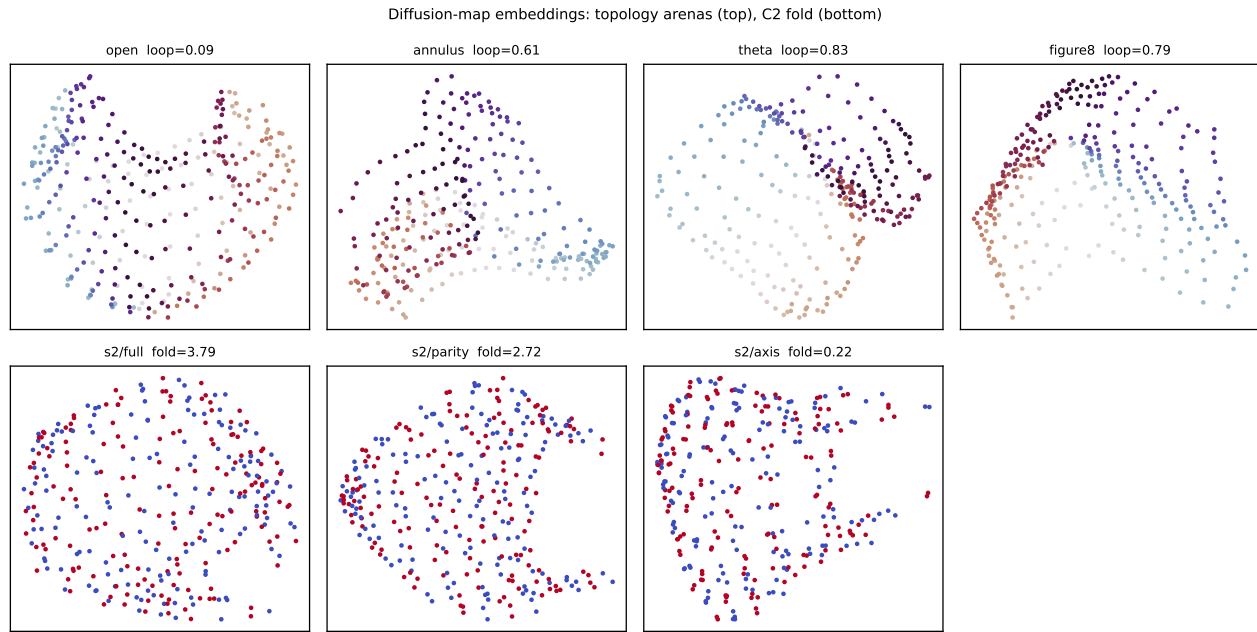


Figure S7. Diffusion-map embeddings. **Top:** topology arenas, coloured by arena angle. The open arena shows no angular loop in the leading diffusion coordinates (loop score 0.09), whereas the annulus, theta and figure-8 arenas do (0.61–0.83); the manifolds nonetheless remain moderately high-dimensional (intrinsic dimension 5–7), consistent with the persistent-homology result that a single loop is recoverable but higher-genus topology is not. **Bottom:** the C_2 fold, coloured by orbit tag. The diffusion distance between a position and its 180° image, in units of the distance to a spatial neighbour, is 3.79 under *full* and 2.72 under *parity* but collapses to 0.22 under *axis*: the fold is a property of the diffusion geometry, not of any linear projection.

5 The wake manifold has the topology of X/G , not X

The previous section's diffusion-distance collapse (position vs. 180° image, 0.22 under *axis* against 2.72–3.79 for the non-invariant encodings) says the fold shows up in a nonlinear, density-robust metric and is not an artefact of a linear projection. It does not by itself say what shape the manifold has taken. We ask that question directly: is the population code, once folded, a map of the physical arena X , or of the quotient X/G ?

If it is X/G , a position and its 180° image under *axis* should not merely be *closer* than average in neural space, they should be nearly the same point, to within the noise the network's own stochasticity sets, while *full* networks (nothing to fold onto) keep them as far apart as any unrelated pair. We tested this on the C_2 arena, across five independently trained networks per encoding: the ratio of neural distance between a position and its 180° image to the neural distance between a random pair of positions.

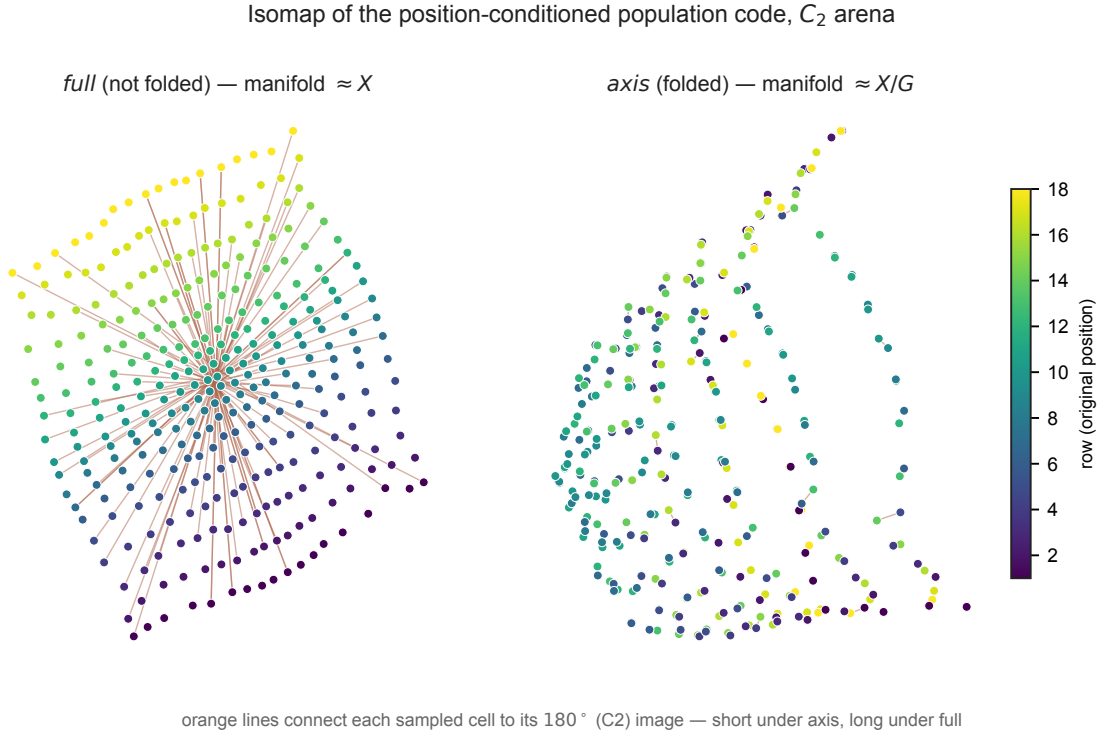


Figure S8. Isomap embedding (geodesic, 2 nearest-neighbour graph, 2 components) of the position-conditioned population code in the C_2 arena, coloured by original row. Lines connect a sample of cells to their 180° image. Under *full* (left) the embedding recovers the arena's own layout and partner lines span the full manifold. Under *axis* (right) partner lines collapse to near zero length: the two symmetry-related halves of the arena have been written onto (approximately) the same manifold, the glued fundamental domain the Theorem predicts, not the unfolded square.

Table S1 gives the ratio for each of the five seeds per encoding. Under *axis* a position sits, on average, 0.17–0.18 of a random pair's distance from its own 180° image; under *full* the same ratio is 0.85–0.89, statistically indistinguishable from a pair with no relationship at all. The separation is complete across all five seeds of each encoding (Mann–Whitney $U = 0$, $p = 0.004$, the exact floor at $n = 5$ vs. 5). This is the topological reading of the quotient law: the network does not merely fail to discriminate 180° -related positions, as the orbit-phase decoding in the main text already shows; it collapses them onto (approximately) the same point of a folded manifold, so that the wake code under a fully G -invariant compass is a map of the fundamental domain, glued, not a map of the arena with two indistinguishable regions on it.

Methods. `analysis/fundamental_domain_isomap.py`. For each of five seeds per encoding, hidden states were collected from the network's own trajectories (20,000 states, matching the orbit-phase decoding protocol) and averaged per arena cell to give a position-conditioned population code (324 cells). The 180° partner of each cell is (19–col, 19–row); the random-pair baseline pairs each cell with an independently shuffled cell, seeded per network. The Isomap embedding

Table S1. Ratio of neural distance between a position and its 180° image to the neural distance between a random pair of positions, C_2 arena, five independently trained networks per encoding.

encoding	seed 0	seed 1	seed 2	seed 3	seed 4
<i>axis</i>	0.172	0.179	0.179	0.167	0.170
<i>full</i>	0.877	0.851	0.869	0.889	0.884

(Fig. S8) uses one representative network per encoding, 8 nearest neighbours, 2 components; the ratio statistics (Table S1) use the raw 500-dimensional hidden state, not the 2-D embedding, so the effect is not a projection artefact. Values in `Report/data/fundamental_domain_isomap.csv`.

6 Initialization and learning speed

The recurrent matrix is not initialized at random alone: the architecture adds a leaky-integrator term $(1 - 1/\tau) I$, and a comment in the original code records that a Kaiming initialization was tried and discarded because it “blows up”. We varied the initialization along three axes — the identity time constant τ , the gain of the random component, and its structure — and summarised each run by the area under its training-loss curve (Fig. S9). Learning speed ordered monotonically with the amount of random recurrent mixing present at initialization. A pure leak, $W = 0.5 I$ with no random recurrence, learned fastest; larger random components learned more slowly, and the two Kaiming variants were slowest of all. Neither Kaiming variant diverged: its loss fell monotonically to within 33% of the default by area, with no sign of instability. The blow-up recorded in the original comment belongs to the plain recurrent cell, not to the layer-normalised cell this network uses. This analysis bears on training dynamics rather than on the symmetry results of the main text, and is reported here for completeness.

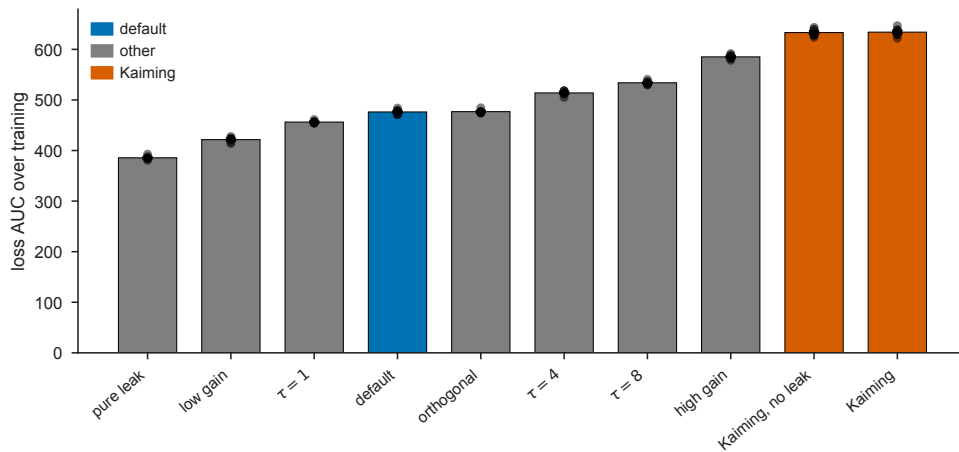


Figure S9. Learning speed by recurrent initialization, measured as area under the training-loss curve (lower is faster), ranked. A pure leak with no random recurrence learns fastest; adding random mixing slows learning monotonically. The abandoned Kaiming initialization is slowest but converges without instability. Dots are individual networks.

7 CA1 reanalysis: exploratory and non-diagnostic

The main text (Fig. 6b) reports a predictive network's own prediction for a city-block maze: same-orientation alleys are translation-related (the compass reads the same) and orthogonal alleys are rotation-related, so a repeating cell's fields should share a directional preference across same-orientation alleys and not across orthogonal ones. Here we ask whether that signature, confirmed in the model, is visible in real neurons.

We reanalysed the CA1 recordings of Hockeimer et al. (2023), in which rats foraged a city-block maze whose alleys run in two orthogonal orientations. Within a repeating cell, the directional index of same-orientation field pairs is positively correlated ($r = 0.23$; 127 pairs, 40 cells, 5 rats; Fig. S10), and pairs in orthogonally-oriented alleys are uncorrelated ($r = -0.06$).

This dataset cannot, however, separate the fold from a conjunctive place-by-direction code, which would produce the same correlation without any folding: a cell with a global directional preference has consistently signed directional indices across all its same-orientation fields. Two controls make the ambiguity explicit. The correlation is if anything stronger among *non-repeating* cells ($r = 0.63$, though only 12 pairs), so it does not require the field repetition the fold predicts; and the directional index is a stable per-cell property (mixed-effects intraclass correlation 0.31), so a cell-scramble that preserves each cell's overall directionality reproduces the correlation ($r = 0.52$, above the observed value). We therefore report the CA1 correlation as *consistent with* the quotient but not diagnostic of it; separating the two accounts requires per-cell head-direction tuning, or the parallel-versus-radial manipulation of Grieves et al. (2016) read out directionally.

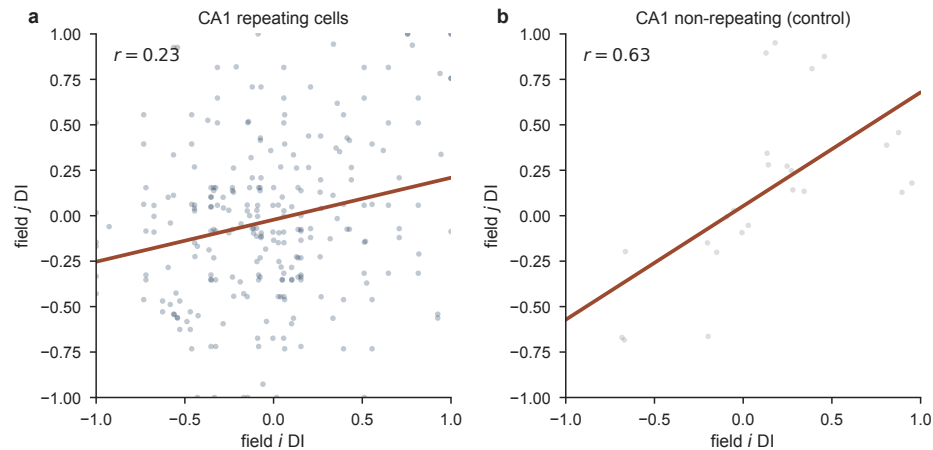


Figure S10. Exploratory reanalysis of CA1 city-block data (Hockeimer et al. 2023): consistent with the quotient but not diagnostic. **(a)** For repeating cells, same-orientation field-DI pairs are weakly positively correlated ($r = 0.23$). **(b)** Non-repeating cells show a similar correlation ($r = 0.63$), so the effect does not require the repetition the fold predicts, and a directionality-matched cell-scramble reproduces it ($r = 0.52$). The correlation is therefore not separable from a conjunctive place-by-direction code; a decisive test needs per-cell head-direction tuning.

8 Coset versus phase: folded, not broken

Figure S11 plots within-domain position (R^2_{domain}) against orbit-phase accuracy for every network and encoding in the main sweep. The two quantities move independently: domain position is high (typically 0.8–0.9) regardless of whether the encoding is invariant, while phase accuracy is high only for the non-invariant encodings and sits at the null for the invariant ones in a symmetric arena. This is the pattern a fold, rather than a collapse of the spatial code altogether, predicts: the network keeps a reproducible map of the fundamental domain and loses only which copy of it produced the current state.

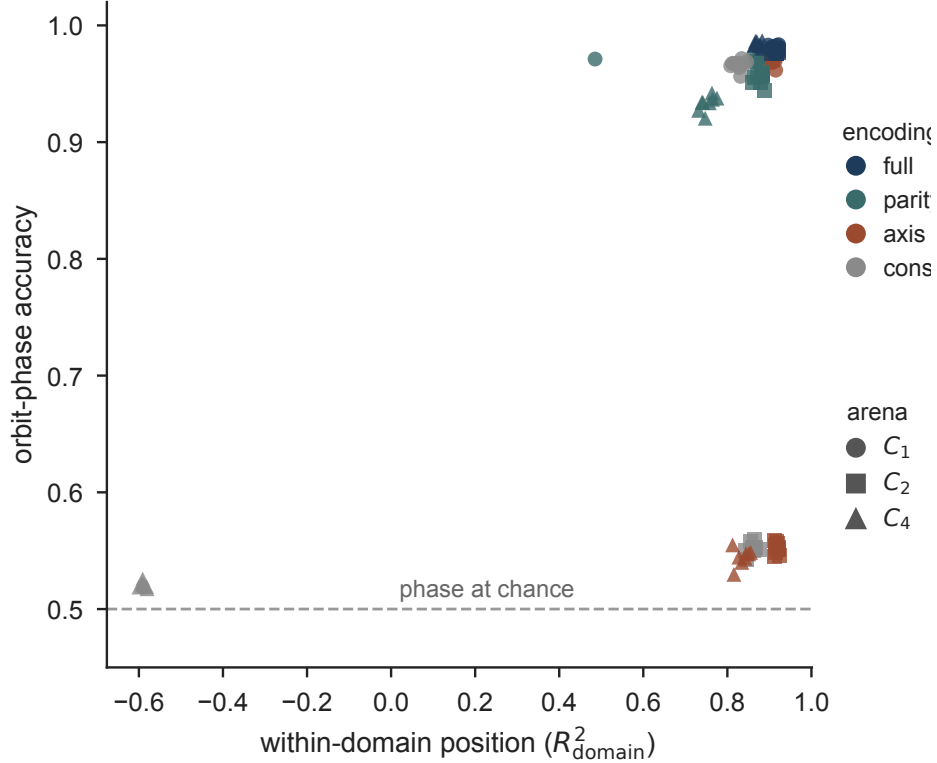


Figure S11. Within-domain position (R^2_{domain}) versus orbit-phase accuracy, every network and encoding, arenas $C_1/C_2/C_4$. High domain R^2 co-occurs with high phase accuracy for the non-invariant encodings (*full*, *parity*) and with chance-level phase accuracy for the invariant ones (*axis*, *const*) in a symmetric arena: the coset is preserved while the phase is not. The point near $R^2_{\text{domain}} = -0.6$ is *const* in the C_4 arena, folded by the full C_4 group in a way a C_2 -based readout cannot see (main text).

9 The fold as a manifold, not just a decoder score: compartment maze

Every fold reported in the main text is a classifier's cross-validated accuracy. Figure S12 asks the same question with no decoder at all: does the shape of the population activity itself look like one room or two? We took the horizon-sweep networks (main text) at $k = 3$ and built an Isomap embedding of the wake population state (cosine metric, 150 nearest neighbours, following Levenstein et al. (2024)), coloured by which physical room produced each state — a label the embedding never sees. For each of the 36 matched positions in the two rooms we measured the distance, in the embedding, between room A's and room B's centroid for that position, normalised by the embedding's overall spread. Under translation, where the Theorem predicts folding, this matched-cell distance is 0.13 ± 0.01 (mean $\pm$ s.d., $n = 8$ seeds): room B's copy of a position embeds almost exactly where room A's does. Under rotation, where the compass turns with the room and the Theorem does not predict folding, the same distance is 1.31 ± 0.16 , an order of magnitude larger (exact two-sided Mann–Whitney U , $n = 8$ vs. 8, $p = 1.55 \times 10^{-4}$, the floor for this n). The result is unchanged embedding into three dimensions instead of two (0.17 ± 0.02 vs. 1.72 ± 0.22 , same p), so the two-dimensional panel below is not compressing structure that only separates in higher dimensions.

The same check at the other completed horizon-sweep values (main text) shows this geometric fold is itself prediction-

dependent, independently of any decoder: at $k = 0$, the autoencoder control with no rollout to predict, translation and rotation are geometrically indistinguishable (0.20 ± 0.01 vs. 0.21 ± 0.03 , $p = 0.23$, not significant). At $k = 1$ the two are already well separated (0.19 ± 0.01 vs. 0.26 ± 0.04 , $p = 1.55 \times 10^{-4}$), and the separation grows substantially larger by $k = 3$ above. This mirrors the main text's decoder-based finding that folding requires a predictive objective, now shown directly in the shape of the population code rather than in a classifier's score.

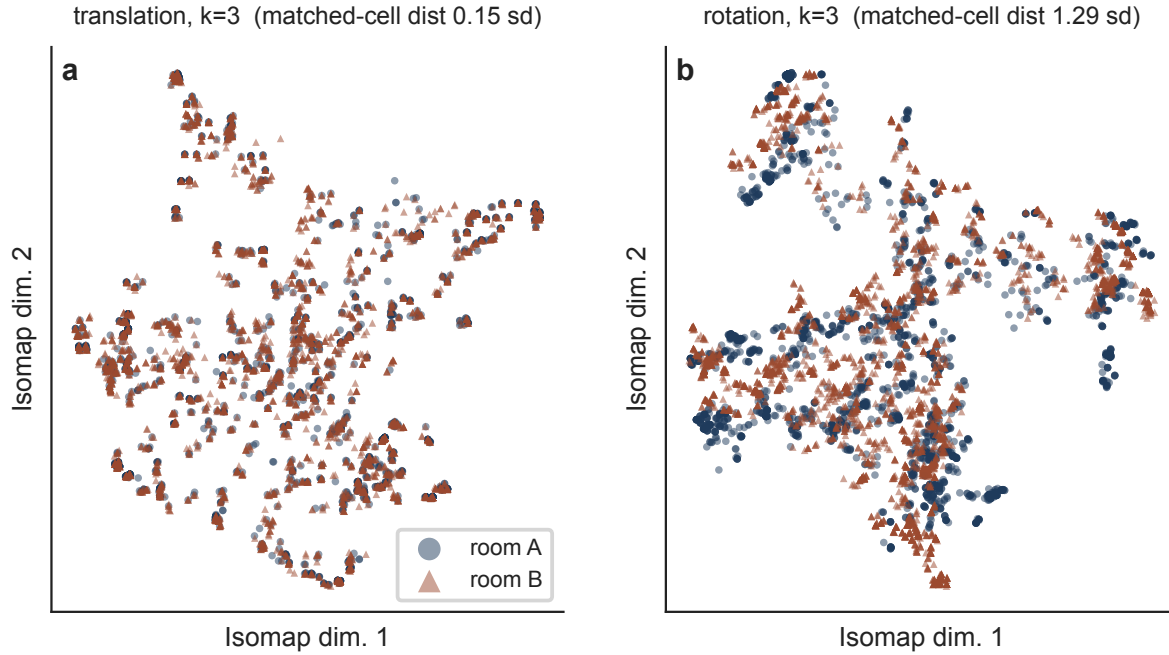


Figure S12. Isomap embedding of the wake population state, compartment maze, $k = 3$, one representative network per condition (the matched-cell-distance statistic in the text is over all 8 seeds). Each point is one moment in time; colour is the room the agent actually occupied (never shown to Isomap). **(a)** Translation: rooms A and B are thoroughly interleaved – the network's internal state cannot tell them apart. **(b)** Rotation: the two rooms occupy visibly more separated regions of the embedding.

10 Nonlinear decoders confirm the chance-level fold

The main text reports orbit phase from a linear (logistic) decoder, so a natural worry is that the folded encodings carry orbit-phase information a linear readout cannot reach. They do not. On the same C_2 population states, a k -nearest-neighbour decoder and a multi-layer perceptron also sit at chance for the invariant encodings (*axis*: 0.537 and 0.535; *const*: 0.531 and 0.531; $n = 6$), and both recover orbit phase for the non-invariant encodings (*full* and *parity*, every network ≥ 0.90). The orbit-phase information is absent from the folded code, not merely hidden from a linear readout.

11 Three single-cell readouts fail in the same way

An invariant compass makes the code look symmetric whether or not the arena is, so single-cell measures of symmetry report a fold even in the C_1 arena, where none can exist. Three readouts we tried early have exactly this defect, which is why the main text reports orbit-phase decoding — whose null survives the C_1 arena — as primary throughout.

Rotational autocorrelation is $RA = P_0 - P_2$ in the isotypic power spectrum (Fig. S13b, c), and is blind to C_2 structure by construction: invariance under R^2 annihilates the odd characters P_1 and P_3 while leaving P_0 and P_2 untouched, so a perfectly C_2 -folded code and pure noise both read $RA \approx 0$. On a clean 17-network sweep, RA cannot separate the C_2 arena from the C_1 one ($p = 0.056$), whereas odd power $P_1 + P_3$ separates them completely ($p = 0.0079$, the exact floor at $n = 5$ vs. 5).

Odd power is itself confounded. Under *axis* it falls in the C_1 arena too, where there is no symmetry: the drop $\text{odd}(\text{parity}) - \text{odd}(\text{axis})$ is 0.095 in C_1 , 0.111 in C_2 and 0.112 in C_4 , so 85% of the C_4 -sized effect is present where nothing can fold, and it separates completely in C_1 ($U = 0$, $p = 1.1 \times 10^{-5}$) — a confident report of a fold that does not exist. It conflates genuine collapse onto the quotient with the encoding merely reshaping the rate-map spectrum.

Spatial representational similarity (the correlation between neural and Euclidean distance) falls as the compass is ablated, from 0.70 under *full* to 0.32 under *const* in the C_1 arena. The fold contributes to this, but is not the main cause: the same fall occurs in C_1 , where nothing folds, and the C_1 drop between *full* and *axis* (0.33) is more than twice the drop the fold itself contributes (the 0.14 between the C_1 and C_2 arenas under *axis*). The rate-map C_2 symmetry index likewise already reads 0.61 in the C_1 arena under *axis* (against the 0.97 it reaches when the map genuinely folds), and the directional tuning curve is already 93% bidirectional there. All three are facts about the compass, not the map; only orbit-phase decoding keeps a null that stays where it should (0.971 in C_1 , 0.552 in C_2 under *axis*).

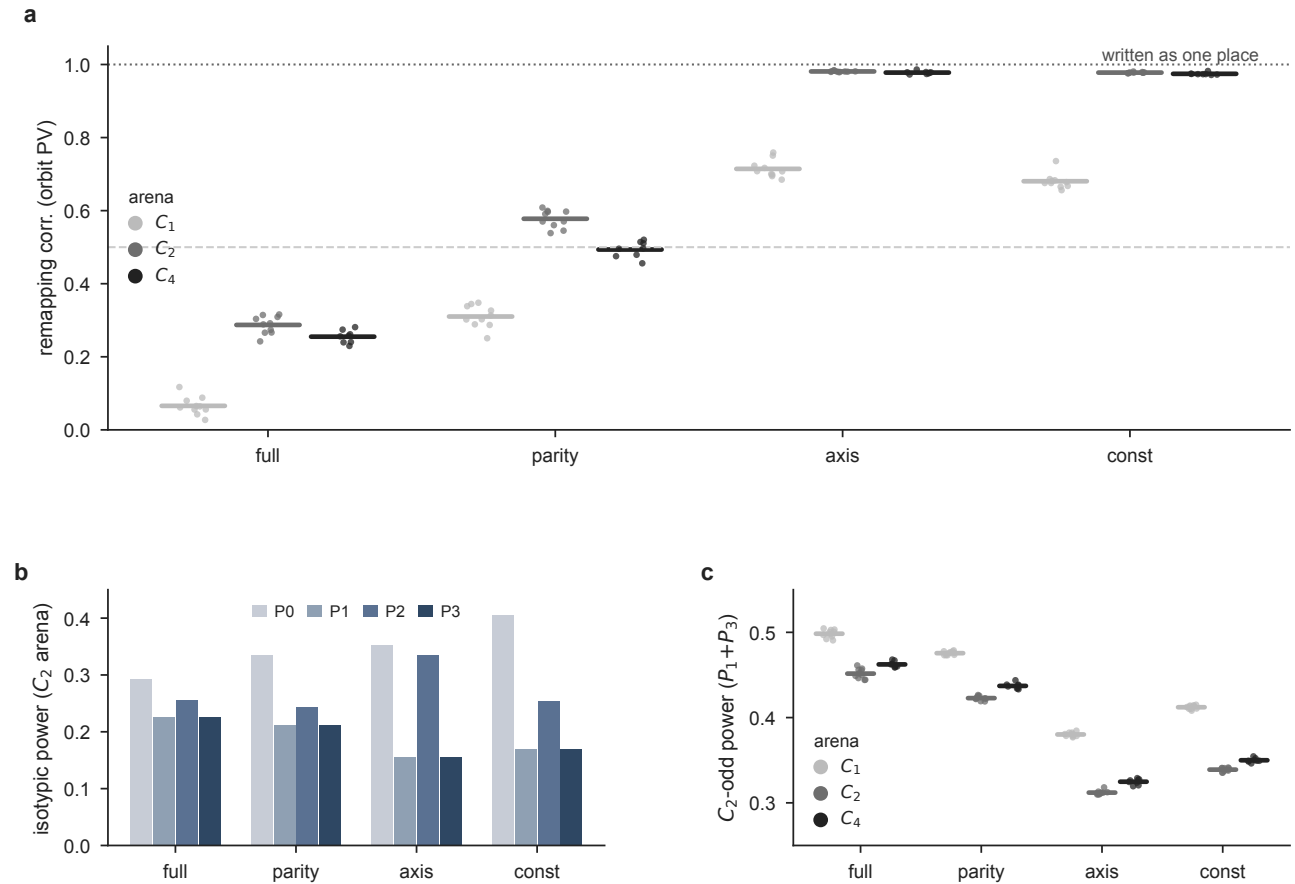


Figure S13. The quotient in the population code. (a) Remapping, read as the population-vector correlation between a position and its symmetry image (Leutgeb et al., 2005) (one dot per network; bar, mean). Under the invariant encodings (*axis*, *const*) the correlation reaches ≈ 0.98 in the C_2 and C_4 arenas, the two copies are written as one place, while *full* separates them (≈ 0.28) and the matched *parity* is intermediate (0.58 in C_2). (b) Isotypic power spectrum (C_2 arena): the invariant encodings concentrate power in the even characters (P_0 , P_2) and suppress the odd ones (P_1 , P_3), the representation-theoretic content the encoding's invariance dictates. (c) C_2 -odd power $P_1 + P_3$ by encoding and arena: lower for the invariant encodings and falling into the symmetric arenas, the descriptive correlate of the fold. Arenas $C_1/C_2/C_4$.

12 The quotient in the population code

Cast in the metrics used to score remapping and directional structure in the hippocampus, the fold appears as a failure to remap between symmetry-related locations exactly where the compass is invariant, and as a suppression of the odd representation-theoretic characters the invariance forbids (Fig. S13).